Master v2 Rev B — PCB design handoff

ESP32-S3 + ESP32-H2 + TCA9548A carrier · 100 × 70 mm · 4-layer

generated from repo — commit current HEAD

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Part 1 — README

Master v2 — ESP32-S3 + ESP32-H2

+ TCA9548A carrier PCB

Rev B. Dedicated controller board for the split-flap display. No onboard flap. Drives up to **8 rows × 16 units = 128 units** via a TCA9548A I2C mux, one I2C bus per row. Adds an ESP32-H2-MINI-1 radio coprocessor for Zigbee 3.0 / Thread / BLE, attached to the primary ESP32-S3 over a UART command/response link.

> **Status**: design-stage, capture-ready. This directory contains handoff artifacts (schematic spec, BOM, pinout, layout constraints). PCB layout is out of scope for this repo — JLCPCB EasyEDA service or a freelance layout engineer takes this brief forward to gerbers.

Related issues

- #59 — this board
- #58 — ESP32-S3 firmware port (`ESPMasterS3/`)
- #55 — power distribution (drives input spec)
- #54 — coil de-energize (per-unit firmware, not PCB)

Rev A → Rev B diff

Area	Rev A	Rev B
Radio coprocessor	—	**ESP32-H2-MINI-1** added (Zigbee/Thread/BLE)
Inter-MCU link	—	**UART1 only** (921600 bps, 8N1, no flow control, command/response)
Buck regulator	MP2315SGJ-Z (SGJ pin 7/8 ambiguity — COMP/SS unclear)	**AP63300WU-7** (TSOT-26, 3.8–32 V V_IN, internally compensated + internal soft-start, fully defined pinout)
Buck compensation	“leave floating” (incorrect — MP2315S needs external RC)	Internal compensation — no COMP/SS network needed
Debug access	S3 native USB + U0 test pads	S3 native USB + **JP_DEBUG_S3** (1×4 UART header) + **JP_DEBUG_H2** (1×6 UART + EN + BOOT_SEL) + H2 USB D±/GND test pads
RF rules	“keep-out under ESP32 antenna”	**Numeric** antenna keep-out (15×7 mm S3, 11×6 mm H2), ≥30 mm antenna edge spacing, opposite-edge placement
BOM	several `VERIFY LCSC` entries	**All LCSC codes committed** — no VERIFY entries
Board size	80 × 60 mm	**100 × 70 mm** (fits second module + antenna clearance)
		4-layer (unchanged

Stackup	4-layer	— justified below)

Compatibility

- **Unit PCBs**: unchanged.
- **Cables**: unchanged. Existing chains plug into the row XH-4 connectors.
- **I2C address space**: unchanged (0x01..0x10 per row, 4-bit DIP).
- **Cap per chain**: 16 units (DIP-switch ceiling).
- **Firmware API**: backwards-compatible on the S3 side; H2 support is additive.

Architecture

- **Primary MCU**: **ESP32-S3-WROOM-1-N16R8** (16 MB flash, 8 MB PSRAM, PCB antenna). Owns system: USB, web UI, WiFi, I2C mux, row scheduler, OTA for both MCUs. Boots autonomously.
- **Radio coprocessor**: **ESP32-H2-MINI-1** (RISC-V, 802.15.4, BLE 5, PCB antenna). Slave to the S3. Power-gated by S3 via `H2_RESET_N` (EN, active-low). Does **not** control system state, I2C, or power.
- **Inter-MCU bus**: **UART only** (hard rule). S3's UART1 ↔ H2's UART1 at 921600 bps, 8N1, no hardware flow control. Framing: COBS-encoded frames with 2-byte CRC-16/CCITT-FALSE footer. Model: S3 is master, H2 is responder. Optional `H2_IRQ` line (H2-asserted, active-low, open-drain with 10 kΩ pullup) signals "data available" so S3 doesn't have to poll. **SPI between MCUs is FORBIDDEN — capture-time check.**
- **I2C mux**: TCA9548APWR @ 3.3 V, address 0x70 (A0/A1/A2 → GND). 8 channels, all used.
- **Per-row level shift**: PCA9306DCTR (VSSOP-8) per row, I2C-specific bidirectional translator.
- **Per-row protection**: 2 A polyfuse (1812L200) + SMBJ5.0A TVS on 5 V + PESD3V3L5UW TVS array on SDA/SCL.
- **Power input**: USB-C (5 V, ≤ 1.5 A — logic only) + 5.5/2.1 mm barrel jack (external PSU, up to 8 A — rows + logic backfeed).
- **Power topology**: dual rail. `5V_LOGIC` (USB via LM66100 U2, jack via D_BACKFEED Schottky) feeds the buck. `5V_RAIL` is fed directly from `JACK_5V` and distributes to rows via polyfuses. LM66100 is on the USB path only; high-current jack current never passes through it.
- **3.3 V rail**: **AP63300WU-7** synchronous buck, 3 A capable, 3.8–32 V input, internally compensated + internal soft-start, 0.8 V reference, fixed 500 kHz. Wider V_IN range chosen to leave ~750 mV margin on jack-only operation (worst-case 5V_LOGIC ≈ 4.55 V after Q1 R_DS(on) + SS14 V_F drops).
- **USB protection**: USBLC6-2SC6 ESD array + ferrite bead on VBUS.
- **Board**: **100 × 70 mm, 4-layer**, controlled impedance on USB diff pair.

Block diagram

```
                  ┌──────────────────────────────────────────────┐
USB-C ──FB1──U1──┤ USB_DP / USB_DN                              │
                  │                                              │
                  │ ┌──────────────────── ESP32-S3-WROOM-1 ────┐ │
                  │ │   GPIO8  I2C SDA  ──── TCA9548A SDA      │ │
                  │ │   GPIO9  I2C SCL  ──── TCA9548A SCL      │ │
                  │ │   GPIO10 MUX_RESET_N — TCA9548A /RESET   │ │
                  │ │   GPIO11 H2_RESET_N  ── H2 EN            │ │
                  │ │   GPIO12 H2_IRQ       ←── H2 GPIO10      │ │
                  │ │   GPIO13 H2_BOOT_SEL ── H2 GPIO9 (strap)│ │
                  │ │   GPIO17 H2_UART_TX  ── H2 GPIO4 (RX)    │ │
                  │ │   GPIO18 H2_UART_RX  ←── H2 GPIO5 (TX)   │ │
                  │ │   GPIO19/20 USB D-/D+                    │ │
```

```
│ │   GPIO43/44 U0TXD/U0RXD ── JP_DEBUG_S3 │ │
│ └──────────────────────────────────────┘ │
│                     │ I2C (3V3)                │
│            ┌────────▼──────────┐                │
│            │    TCA9548A (U6)   │                │
│            │   8× mux channels  │                │
│            └────────────────────┘                │
│                 │   │      │                     │
│            ┌────▼───▼──────▼────┐                │
│            │  8× PCA9306 (U7..)│                │
│            │  3V3 ↔ 5V level    │                │
│            │     translators    │                │
│            └────────────────────┘                │
│                 │   │      │                     │
│           ROW1 ROW2 .. ROW8 (XH-4)               │
│                                                 │
│ ┌──────────── ESP32-H2-MINI-1 ────────────┐   │
│ │   Zigbee 3.0 / Thread / BLE 5           │ │
│ │   UART1 ↔ S3 (921600 8N1, command/rsp) │ │
│ │   GPIO23/24 U0TXD/U0RXD → JP_DEBUG_H2   │ │
│ │   GPIO26/27 USB D-/D+ → test pads       │ │
│ └──────────────────────────────────────┘   │
└──────────────────────────────────────────────┘
```

```
Power tree:
  USB-C VBUS ── FB1 ── U2 LM66100 ── 5V_LOGIC ──┬── U4 AP63300 buck
── 3V3_RAIL
                                                │
                                   D_BACKFEED SS14 (anode on
JACK_5V)
                                                │
  Barrel jack ── Q1 ── D1 ── JACK_5V ──────────┴──────── direct
──────── 5V_RAIL ──┬── F1..F8 ── ROW1..8

│
                                                        per row:
PCA9306 + PESD + SMBJ

L2 GND plane uninterrupted. L3 = 5V_RAIL pour + 3V3 island under
MCUs.
```

Key specs

Item	Value
Primary MCU	ESP32-S3-WROOM-1-N16R8
Radio coprocessor	ESP32-H2-MINI-1-N4
Inter-MCU link	UART1, 921600 bps, 8N1, no flow control, COBS + CRC-16
I2C mux	TCA9548APWR @ 3.3 V, addr 0x70
Rows	8 × XH-4 (5 V / GND / SDA / SCL)
Level shift	PCA9306DCTR (VSSOP-8) per row
Per-row protection	2 A polyfuse + SMBJ5.0A TVS + PESD3V3L5UW
Power input	USB-C 5 V (logic) + 5.5/2.1 mm barrel jack 5 V (rows + logic backfeed)
Logic OR-ing	LM66100DCKR × 1 (USB path only, **not** in high-current path)
Jack-to-logic backfeed	SS14 Schottky, 1 A
3.3 V	AP63300WU-7 synchronous buck, 3 A, 3.8–32 V V_IN, internally compensated, 500 kHz

Rails	5V_LOGIC (buck input) + 5V_RAIL (rows) — distinct nets
USB ESD	USBLC6-2SC6 + BLM18PG471 ferrite
Buttons	BOOT_S3 (S3 GPIO0), RESET_S3 (S3 EN)
LEDs	Power (always-on), WiFi (S3 GPIO4), Activity (S3 GPIO5), Zigbee (S3 GPIO6)
Debug	Native USB (S3) + JP_DEBUG_S3 1×4 UART + JP_DEBUG_H2 1×6 UART+EN+BOOT + H2 USB D± test pads
Board	100 × 70 mm, 4-layer, 1.6 mm FR4
Input current ceiling	8 A total (input spec)

Bulk capacitance summary

Net	Total bulk	Breakdown
JACK_5V (input)	**480 µF**	470 µF Al-polymer + 10 µF X7R
5V_RAIL (row)	**267 µF + HF**	220 µF Al-polymer + 47 µF X5R + 100 nF
5V_LOGIC (buck input)	**32 µF**	22 µF X5R (buck C_IN) + 10 µF X7R (U2 OUT)
3V3_RAIL (logic)	**76 µF**	2× 22 µF X5R (buck out) + 22 µF X5R (S3 bulk) + 10 µF X7R (H2 bulk) + local HF pairs
Per-row (at connector)	**≥ 47 µF + 100 nF**	C_ROW{n} 47 µF X5R 1210 + C_ROW{n}_HF 100 nF 0603

Per-row capacitance meets the ≥ 47 µF hard requirement.

Stackup — 4-layer (decision + justification)

Choice: 4-layer. Rejected 6-layer.

Justification: - Only two positive power rails (5 V, 3.3 V) + GND. 6-layer adds a second GND or a third power layer that this design does not need. - No high-speed DDR/HBM interconnect — PSRAM is integrated inside the S3 module. - Single controlled-impedance pair (USB D+/D−) routes cleanly on L1 over an uninterrupted L2 GND. - Two 2.4 GHz antennas — both module-integrated with vendor-validated keep-outs. No on-board RF traces requiring stripline. - Density: S3 module + H2 module + TCA9548A + 8× PCA9306 + connectors + passives fits comfortably on L1 (top) with spillover to L4 (bottom). L3 power-plane splits (5V_RAIL / 3V3 island) are the only routing constraint on the inner layer and do not require a second signal layer. - Cost: 6-layer roughly doubles fabrication cost at JLCPCB for no measurable benefit here.

Stackup (JLCPCB `JLC04161H-7628` or equivalent, 1.6 mm FR4, Tg 155):

Layer	Copper	Function
L1 (top)	1 oz	**Signals + components.** All IC placement. USB diff pair. I2C trunk. Local decoupling routing. SW node for buck.
L2 (inner 1)	0.5 oz	**Solid GND plane.** Uninterrupted — no splits, no cutouts (except antenna keep-outs). Primary return reference for USB diff pair and buck SW node.
L3 (inner 2)	0.5 oz	**Power distribution.** 5V_RAIL solid pour (row power). Carved 3V3 island under both MCU modules + mux + PCA9306 VREF1 pins. 5V_LOGIC sub-island around buck input.
L4 (bottom)	1 oz	**Secondary signals + GND stitch.** UART link between MCUs, miscellaneous control signals, H2 debug breakout. Unused areas flooded GND, stitched to L2 on 5 mm grid.

Dielectric: 0.3 mm prepreg L1–L2, 1.065 mm core L2–L3, 0.3 mm prepreg L3–L4. Dk 4.4 @ 1 GHz.

Controlled impedance: **90 Ω differential** on L1 over L2 for USB D+/D−. Trace/space 0.18 / 0.13 mm. Specify impedance test coupon on fabrication order.

RF constraints (numeric, mandatory)

Both modules carry their own PCB antennas and operate in the 2.4 GHz ISM band. **Vague wording is a FAIL condition. All values below are hard rules.**

ESP32-S3-WROOM-1 antenna

- Keep-out region: **15 mm × 7 mm** from the antenna end of the module.
- **No copper** on any of the 4 layers inside the keep-out — L1, L2, L3, L4 all cleared. This breaks the L2 GND plane locally; stitching must route *around* the keep-out, not through.
- **No traces** routed under the keep-out on any layer.
- **No ground flooding** inside the keep-out on any layer.
- Module orientation: antenna edge faces **PCB top edge** (long side of board).

ESP32-H2-MINI-1 antenna

- Keep-out region: **11 mm × 6 mm** from the antenna end of the module.
- Same rules as S3 keep-out: no copper on any layer, no traces underneath, no flood.

- Module orientation: antenna edge faces **PCB bottom edge** (opposite edge from S3 antenna) — maximum physical separation.

Inter-module RF isolation

- **Minimum antenna-to-antenna edge distance: 30 mm.** Measured between the nearest points of the two keep-out regions.
- Modules on opposite board edges (S3 top, H2 bottom): ~60 mm achievable on a 70 mm-tall board. Exceeds the 30 mm floor by 2×.
- No common-mode coupling path: both modules share GND on L2 (required) but with uninterrupted plane between them.
- **Coexistence note** (firmware, not PCB): S3 WiFi/BT and H2 802.15.4/BLE both occupy 2.4 GHz. Hardware does not arbitrate. Firmware MUST implement TDM — typical approach: S3 operates on WiFi channel 1 (2401–2423 MHz), H2 pinned to 802.15.4 channel 25 or 26 (2475–2480 MHz) to avoid spectral overlap. This is an application-layer constraint; the PCB does not preclude it.

Antenna placement — REQUIRED orientation

```
┌──────────────────── top edge ────────────────────┐
│  ▓▓▓▓▓▓▓▓  ← S3 antenna keep-out (15×7)           │
│  ┌──────────┐                                    │
│  │  ESP32-S3  │                                  │
│  └──────────┘                                    │
│                                                  │
│     (remainder of components)                    │
│                                                  │
│  ┌──────────┐                                    │
│  │ ESP32-H2  │                                   │
│  └──────────┘                                    │
│  ▓▓▓▓▓▓  ← H2 antenna keep-out (11×6)            │
└──────────────── bottom edge ─────────────────────┘
```

Firmware constraints

The board electrically allows illegal operating modes; firmware MUST enforce:

- **Single-channel-at-a-time mux selection** (TCA9548A). Channel-select register writes must have exactly one bit set: 0x01, 0x02, 0x04, 0x08, 0x10, 0x20, 0x40, or 0x80. Multi-bit masks parallel-combine per-row pullups and violate ATmega328P I_OL (3 mA max).
- **Pre-flight mux select guard** rejects non-power-of-two values.
- **MUX_RESET_N pulse** (≥ 10 µs low) during I2C bus-stuck recovery.
- Bank-aware probe: iterate (bank 0..7, address 0x01..0x10) = 128 slots max.
- **H2 power sequencing**: on S3 boot, hold H2_RESET_N low for ≥ 10 ms, ensure H2_BOOT_SEL high (normal SPI boot), then release H2_RESET_N. Wait ≥ 100 ms for H2 firmware to start. First UART transaction is a hello/version handshake.
- **H2 OTA**: S3 is the authoritative firmware provider. To reflash H2, S3 pulls H2_BOOT_SEL low, pulses H2_RESET_N, and drives the esptool-compatible UART protocol at 921600 bps to write new H2 firmware to H2's SPI flash. Then S3 releases H2_BOOT_SEL and H2_RESET_N for normal boot.
- **UART arbitration**: S3 is always master. H2 only transmits in response to a command, or when it asserts H2_IRQ first and S3 replies with a POLL command.
- **2.4 GHz coexistence**: at minimum, pin H2 to 802.15.4 channel 25 or 26 and S3 WiFi to channel 1. Do not run H2 BLE scanning concurrently with WiFi high-throughput transfers.

Bill of materials

BOM.csv. JLCPCB PCBA-ready. **All entries have committed LCSC part numbers — no VERIFY stubs.** LCSC stock is order-time; substitutions (same part, different manufacturer) are acceptable if package and electricals match.

Schematic

- SCHEMATIC.md — authoritative textual netlist.
- master_v2.kicad_sch — minimal KiCad container. Layout engineer recaptures from SCHEMATIC.md.

Pinout

PINOUT.md — full ESP32-S3 and ESP32-H2 GPIO tables, strap-pin rules, reserved pins.

Handoff workflow (mandatory, sequential)

1. **Schematic capture** in target EDA from SCHEMATIC.md + BOM.csv. ERC must pass with zero errors and zero warnings.
2. **Pre-layout review checklist** (all 8 signed off before routing starts):
   1. PCA9306DCTR pin numbering verified against TI SCPS169 Table 3 (VSSOP-8: 1 EN, 2 VREF1, 3 SDA1, 4 GND, 5 SCL1, 6 SCL2, 7 VREF2, 8 SDA2).
   2. TCA9548APWR pin numbering verified against TI SCPS195 (pin 1 A0, pin 4 /RESET, pin 12 GND, pin 22 SCL, pin 23 SDA, pin 24 VCC; channel pin mapping per Table 5-1).
   3. USBLC6-2SC6 routing confirmed: pins 1+6 = IO1 (tie to one D-line), pins 3+4 = IO2 (tie to other D-line), pin 5 = VBUS, pin 2 = GND. **Pin 4 is NOT ground.**
   4. AP63300WU-7 feedback divider: R_FB_TOP 100 kΩ + R_FB_BOT 31.6 kΩ → V_OUT 3.34 V ±1 % (V_FB = 0.803 V typical). C_FB 22 pF present across R_FB_TOP. C_BOOT 100 nF between pins 6 (BST) and 1 (SW).
   5. ESP32-S3 strap pins: GPIO0 drives only BOOT_S3_BTN. GPIO3, GPIO45, GPIO46 have no copper escape beyond module pad.
   6. ESP32-H2-MINI-1 strap pins: IO2, IO8 tied high via 10 kΩ to 3V3. IO9 pulled high via 10 kΩ to 3V3 with 1 kΩ series from S3 GPIO13 for optional pull-low (download mode).
   7. Inter-MCU link is **UART only**. No SPI nets between S3 and H2. No shared I2C. ERC must confirm `H2_UART_TX`, `H2_UART_RX`, `H2_RESET_N`, `H2_IRQ`, `H2_BOOT_SEL` are the only signal nets crossing between modules.
   8. Antenna keep-outs (S3 15×7 mm, H2 11×6 mm) clear on all 4 layers; modules oriented on opposite board edges with ≥ 30 mm antenna edge-to-edge separation.
3. **Placement review** against §“Layout constraints” below.
4. **Routing review** against §“Routing constraints” below.
5. **DRC + impedance report** delivered with gerbers.

Layout constraints

Grounding

- L2 is the single GND reference. Uninterrupted, no splits.
- Every IC GND pin stitches to L2 with ≥ 2 vias.
- L4 bottom pour is GND, stitched to L2 on a 5 mm grid via array.
- Antenna keep-outs (15 × 7 mm under S3 antenna, 11 × 6 mm under H2 antenna): no copper on any of the 4 layers.

High-current return

- Row connector GND pins → direct line-of-sight through L2 back to barrel-jack GND lug and USB-C shield.
- No row-return currents route beneath either MCU module or the antenna keep-outs.

5 V distribution (dual rail)

- **5V_RAIL** (row power): L3 solid power pour, fed from JACK_5V direct. Polyfuses F1..F8 break the pour into per-row outbound islands.
- **5V_LOGIC** (buck input): separate L3 island or dedicated L1 trace ≥ 1.0 mm wide, fed by U2 LM66100 output and D_BACKFEED cathode. Does not connect to 5V_RAIL anywhere on the board except via D_BACKFEED Schottky (anode side on JACK_5V).
- **3V3 island** on L3 beneath both MCU modules + TCA9548A + PCA9306 VREF1 pins; fed from buck output via ≥ 1.5 mm L1 trace through two vias.

Buck (U4 AP63300WU-7 + L_BUCK)

- C_BUCK_IN within 2 mm of U4 VIN pin on 5V_LOGIC net.
- L_BUCK within 4 mm of U4 SW pin.
- SW trace: L1 only, ≤ 6 mm length, ≤ 3 mm² area.
- L2 GND under SW node uninterrupted (no L2 cutouts under U4).
- C_BUCK_OUT_1 within 3 mm of L_BUCK output; C_BUCK_OUT_2 within 5 mm.
- Feedback divider within 5 mm of U4 FB pin; FB trace flanked by GND pour on L1.
- No signal routing under L_BUCK or under SW node on any layer.
- U4 SW pin / thermal copper: ≥ 50 mm² copper area on L1 connected to the SW pad plus GND pour. Stitch vias (≥ 6 × 0.3 mm) from top-layer GND pour under U4 down to L2.

USB and USB-C connector

- USBLC6-2SC6 placement: **≤ 3 mm from USB-C D+/D− pins**. No trace reaches the ESP32-S3 side before passing U1.
- D+/D− differential pair: 90 Ω on L1 referenced to L2, width 0.18 mm, spacing 0.13 mm, length-matched ±0.2 mm, max length 50 mm from U1 to ESP32-S3 GPIO19/20.
- VBUS ferrite bead FB1 within 5 mm of connector VBUS pin.
- Cable shield tied to GND through a 1 MΩ resistor in parallel with a 10 nF cap (chassis path).

Inter-MCU UART

- `H2_UART_TX`, `H2_UART_RX`, `H2_RESET_N`, `H2_IRQ`, `H2_BOOT_SEL`: route primarily on L4 referenced to L2 GND. Max length 80 mm (corner-to-corner on 100 × 70 mm board).
- Keep ≥ 5 mm lateral clearance from the buck SW node trace and ≥ 3 mm from the USB diff pair.
- Series damping resistors (R_H2_RST_SER 1 kΩ, R_H2_BOOT_SER 1 kΩ) on S3 side, within 5 mm of S3 module pad.

I2C routing

- ESP32-S3 → TCA9548A trunk (SDA/SCL on L1): ≤ 30 mm.
- TCA9548A channel n → PCA9306 U(6+n) 3V3 side: ≤ 40 mm on L1.
- PCA9306 5V side → XH-4 connector pin: ≤ 15 mm on L1.
- Bypass jumpers JP_BYPASS_SDA / JP_BYPASS_SCL (0 Ω DNP) between ESP32-S3 bus and TCA9548A channel 0 outputs.

Per-row protection

- TVS D_ROW{n} within 5 mm of J{n} pin 1 (5 V).
- PESD3V3L5UW within 5 mm of J{n} pins 3/4 (SDA/SCL).

- C_ROW{n} (47 µF 1210) and C_ROW{n}_HF (100 nF 0603) within 5 mm of J{n} power pins.
- Polyfuse F{n} in series in the J{n} 5 V path; ≥ 1.0 mm L1 bridge to L3 island.

Strap pins and module

- S3 GPIO0 trace to BOOT_S3_BTN ≤ 20 mm, flanked by GND pour.
- S3 GPIO3, GPIO45, GPIO46: no copper escape beyond module pad on any layer.
- H2 IO2, IO8: 10 kΩ pullup to 3V3 within 5 mm of module pad.
- H2 IO9: 10 kΩ pullup to 3V3 within 5 mm of module pad, series 1 kΩ from S3 GPIO13.

Test points

- 1.0 mm exposed round pad, silkscreen labels per PINOUT.md and SCHEMATIC.md §test-points.
- TP_MUX_SDA0..7 and TP_MUX_SCL0..7 arranged in a linear row adjacent to TCA9548A for scope probe access.
- TP_H2_USB_DP / TP_H2_USB_DN: 1 mm pads near H2 module for optional USB-JTAG access (wire-soldered pigtail or pogo jig).

DRC

- Min trace/space: 0.15 / 0.15 mm.
- Via: 0.3 mm drill / 0.6 mm pad.
- Minimum annular ring: 0.05 mm.

Manufacturing

- Service: JLCPCB PCBA, 4-layer FR4 Tg 155, ENIG finish.
- Assembly: top side only. Through-hole connectors (barrel jack, XH-4, pin headers) hand-solder or JLCPCB extended-THT service.
- Controlled impedance: request 90 Ω differential test coupon with the fabrication order.

Bring-up checklist

Pre-power: 1. Visual inspection — solder bridges, polarity of TVS, TCA9548A pin 1, PCA9306 pin 1, S3 module pin 1, H2 module pin 1. 2. Continuity: USB VBUS ↔ GND open; 5V_RAIL ↔ GND open; 3V3_RAIL ↔ GND open. 3. Barrel-jack reverse polarity: apply reversed 5 V, confirm no current draw (PMOS Q1 off). 4. JP_BYPASS_SDA, JP_BYPASS_SCL DNP — unpopulated for first power-on.

Power-up, no firmware: 1. Apply 5 V barrel. Measure at TP_5V = 5.0 V ±5 %. Measure at TP_3V3 = 3.33 V ±3 %. 2. Scope U4 SW node: ripple ≤ 150 mV pk-pk at ~500 kHz (AP63300 fixed frequency). 3. Measure 5 V at each row connector pin 1 under no load. 4. USB-C only: confirm CDC device enumerates on host.

Firmware bring-up: 1. Flash blinky on S3 GPIO4. Confirm blink. 2. I2C scan at TP_MCU_SDA/SCL returns 0x70 (TCA9548A), no other address with rows unplugged. 3. Probe JP_DEBUG with logic analyzer — transaction patterns match ESP32 I2C API. 4. For each bank (0..7), single-channel select, scan. Probe TP_MUX_SDA{n}/SCL{n} (3.3 V logic) and TP_ROW{n}_SDA/SCL (5 V logic). Both show identical ACK pattern; rise time ≤ 1 µs. 5. H2 bring-up: flash H2 factory image via JP_DEBUG_H2 with external USB-UART dongle (3V3, GND, U0TXD, U0RXD, EN, BOOT_SEL). Confirm H2 boots and prints ROM banner on U0TXD at 115200 8N1. 6. S3 ↔ H2 link: run handshake probe. S3 asserts H2_RESET_N low 10 ms, releases, waits 100 ms, sends `{0x7E`,

`0x01, 0x00, CRC_H, CRC_L, 0x00}` (COBS-framed PING). H2 replies with PONG frame. TP on S3 GPIO17/18 shows 921600 bps traffic in both directions.

Hardware validation: 1. Hot-unplug a row cable under 100 mA load. Scope TP_ROW{n}_5V: clamp peak ≤ 9.2 V (SMBJ5.0A spec). 2. Thermal soak U4 at 500 mA load, 10 min. U4 package ≤ 85 °C at 40 °C ambient (IR probe). 3. Simultaneous 16-unit step command on one row. TP_5V droop ≤ 200 mV pk, recovery ≤ 2 ms. Droop > 500 mV indicates bulk cap ESR out of spec. 4. Firmware guard test: writing 0x03 (two-channel enable) to mux is refused by firmware; scope TP_MCU_SDA confirms no slave ACK to illegal write. 5. H2 radio: run Zigbee scan on H2, S3 relays results over UART, log to S3 serial. Confirm neighbour PAN discovery. 6. Coexistence: run WiFi iperf on S3 concurrent with Zigbee ping on H2. Both must sustain traffic; monitor for packet loss correlation.

Mux-bypass diagnostic (only when diagnosing mux failure): 1. Power off. Populate JP_BYPASS_SDA and JP_BYPASS_SCL (0 Ω). 2. Pull MUX_RESET_N low via test clip (holds TCA9548A in reset / high-Z). 3. Plug chain into ROW1 only. Power on. 4. ESP32-S3 I2C scan at TP_ROW1_SDA/SCL returns 0x01..0x10. Validates MCU → PCA9306 → unit chain independent of mux. 5. Restore: remove jumpers, release MUX_RESET_N.

Part 2 — SCHEMATIC

Master v2 — textual schematic

Rev B. Authoritative netlist. Layout engineer recaptures in target EDA (KiCad 8 or EasyEDA) from this file + BOM.csv.

Conventions: - Net names in `UPPER_SNAKE_CASE`. - Designators match BOM.csv. - "→" means "connects to". - Passive values inline. - Every pin of every IC is defined. Every net has a name. No "verify later" / "approximate" / "representative" language.

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1. USB-C input (J1, 16-pin receptacle, USB 2.0)

USB-C pin(s)	Net	Notes
A4, A9, B4, B9	**USB_VBUS**	VBUS, tied internally
A1, A12, B1, B12, shield	**GND**	including cable shield
A5	**USB_CC1**	5.1 kΩ R_CC1 → GND
B5	**USB_CC2**	5.1 kΩ R_CC2 → GND
A6, B6	**USB_DP_RAW**	to USBLC6 U1
A7, B7	**USB_DN_RAW**	to USBLC6 U1
A2, A3, A10, A11, B2, B3, B10, B11	NC	USB 2.0 only, SuperSpeed pairs not used

- USB_VBUS → ferrite bead **FB1** (BLM18PG471SN1D, 470 Ω @ 100 MHz) → **USB_5V_IN**
- Cable shield bonded to GND through 1 MΩ **R_SH1** in parallel with 10 nF **C_SH1**.

2. USB ESD protection — U1 USBLC6-2SC6 (SOT-23-6)

Pin	Signal	Net
1	I/O 1	**USB_DN_RAW ↔ USB_DN** (tied to pin 6)
2	GND	**GND**
3	I/O 2	**USB_DP_RAW ↔ USB_DP** (tied to pin 4)
4	I/O 2	tied to pin 3 — **NOT GND**
5	VBUS	**USB_5V_IN**
6	I/O 1	tied to pin 1

- USB_DN → ESP32-S3 GPIO19
- USB_DP → ESP32-S3 GPIO20

3. Barrel jack (J2, DC-005 5.5/2.1 mm, center-positive)

- Center pin → **JACK_IN_RAW**
- Sleeve → **GND**

Reverse polarity — Q1 AO3401A PMOS (SOT-23-3)

- S (pin 3) → **JACK_IN_RAW**
- G (pin 1) → **GND** via 10 kΩ **R_QGATE**
- D (pin 2) → **JACK_5V**

TVS — D1 SMAJ5.0CA (DO-214AC)

- Between **JACK_5V** and **GND**, bidirectional, 5 V working, 9.2 V clamp.

4. Power path (dual-rail split)

Two independent use cases:

1. **USB input** — low current (≤ 1.5 A), powers logic only.
2. **Barrel jack input** — high current (up to 8 A system load), powers row 5 V directly plus a low-current backfeed for logic.

Two distinct power nets:

- **5V_LOGIC** — buck converter input + any low-current 5 V loads.
- **5V_RAIL** — row polyfuses + per-row 5 V distribution.

USB path (protected, limited current) — U2 LM66100DCKR (SOT-23-5)

Pin	Name	Net
1	IN	**USB_5V_IN**
2	GND	**GND**
3	ST (status)	NC
4	/CE (enable, active low)	pulled to IN via 100 kΩ **R_CE_USB** (always enabled)
5	OUT	**5V_LOGIC**

Purpose: safe OR-ing for USB-powered bench operation, ~1.5 A max. Does not feed row power.

Barrel jack path (HIGH CURRENT, BYPASSES LM66100)

- **JACK_5V** (post Q1, D1) → **5V_RAIL** directly through polyfuse distribution only. No LM66100 in this path.
- Polyfuses F1..F8 (§10) break 5V_RAIL into per-row segments.

CRITICAL: JACK_5V MUST NOT pass through any LM66100. Current up to 8 A destroys the device.

Logic backfeed from jack (low current)

- **D_BACKFEED** Schottky (SS14, 40 V, 1 A, V_F ≤ 0.4 V, DO-214AC): anode on **JACK_5V**, cathode on **5V_LOGIC**.
- Jack-only operation: logic runs at ~4.55 V (well within AP63300 V_IN 3.8–32 V range, ~750 mV margin).
- With USB present: U2 asserts 5.0 V on 5V_LOGIC; D_BACKFEED reverse-biased.
- Backfeed current capped by buck input demand (≤ 600 mA).

Bulk capacitance

Cap	Value	Net	Placement
C_BULK_INPUT	470 µF 10 V Al-polymer, ESR ≤ 25 mΩ	JACK_5V → GND	near Q1 output
C_BULK_5V_RAIL	220 µF 10 V Al-polymer, ESR ≤ 30 mΩ	5V_RAIL → GND	central, between row fuses
C_BULK_5V_CER	47 µF 10 V X5R 1210	5V_RAIL → GND	central
C_BULK_5V_HF	100 nF 0603 X7R	5V_RAIL → GND	central
C_USB_IN	10 µF 25 V X7R 0805	USB_5V_IN → GND	near U1/U2
C_JACK_IN	10 µF 25 V X7R 0805	JACK_5V → GND	near Q1 output
C_LOGIC_IN	10 µF 25 V X7R 0805	5V_LOGIC → GND	near U2 output

Per-row caps: see §10.

5. 3.3 V regulator — U4 AP63300WU-7 synchronous buck (TSOT-26)

Part: **AP63300WU-7** (Diodes Incorporated). 3.8 V – 32 V input, 3 A, integrated synchronous MOSFETs, internally compensated, 0.803 V feedback reference, **fixed 500 kHz** switching. Internal compensation and internal soft-start — no external COMP or SS network required.

Input range 3.8 V chosen over TPS54302's 4.5 V specifically to tolerate worst-case jack-only operation: 5.0 V source − AO3401A R_DS(on) drop (~50 mV @ 500 mA) − SS14 V_F (~0.4 V @ 500 mA) = ~4.55 V on 5V_LOGIC. AP63300 has ~750 mV headroom above its 3.8 V minimum; TPS54302 had only ~50 mV above its 4.5 V minimum.

U4 AP63300WU-7 pinout (TSOT-26)

Pin	Name	Net
1	SW	**SW_NODE** — to L_BUCK input; also bootstrap capacitor

		return
2	GND	**GND**
3	FB	**FB_NODE**
4	EN	tied to **5V_LOGIC** via 100 kΩ **R_EN_BUCK** (always enabled)
5	VIN	**5V_LOGIC**
6	BST	**BST_NODE** — via **C_BOOT** 100 nF 0603 X7R to SW (pin 1)

External network (required)

- **L_BUCK** 4.7 µH, shielded, I_SAT ≥ 3 A, DCR ≤ 50 mΩ: between **SW_NODE** and **3V3_RAIL**.
- **C_BUCK_IN** 22 µF 25 V X5R 1210: **5V_LOGIC** (U4 pin 5) → GND.
- **C_BUCK_OUT_1** 22 µF 25 V X5R 0805: **3V3_RAIL** (L_BUCK output) → GND.
- **C_BUCK_OUT_2** 22 µF 25 V X5R 0805: **3V3_RAIL** → GND. Total output capacitance ≥ 44 µF.
- **R_FB_TOP** 100 kΩ 1 % 0603: **3V3_RAIL** → **FB_NODE**.
- **R_FB_BOT** 31.6 kΩ 1 % 0603: **FB_NODE** → GND.
- **C_FB** 22 pF 0603 C0G: across R_FB_TOP (feedforward zero).
- **C_BOOT** 100 nF 0603 X7R: between pin 6 (BST) and SW node (pin 1).

V_OUT = 0.803 V × (1 + 100 kΩ / 31.6 kΩ) = **3.34 V** (nominal 3.3 V ±1 % typical).

Layout requirements (MANDATORY)

- L_BUCK and C_BUCK_OUT_1 within 4 mm of U4.
- SW node copper area ≤ 3 mm².
- SW trace on L1 only; L2 GND plane directly beneath, uninterrupted.
- Feedback divider within 5 mm of U4 pin 3; FB trace flanked by GND pour.
- BST capacitor within 3 mm of U4 between pins 6 and 1.
- No signal routing under L_BUCK or under SW node on any layer.

Thermal

- AP63300WU-7 is TSOT-26 with no explicit EPAD. Use ≥ 50 mm² copper area on L1 connected to GND pin (pin 2) and SW pin (pin 1) pour, with ≥ 6 stitching vias (0.3 mm drill) to L2 under the package footprint. Package θ_JA with recommended copper area is ~80 °C/W per Diodes datasheet — at 3 A × (1 − 0.66 efficiency) × V_IN ≈ 1.7 W dissipation worst case, rise ≈ 136 °C is over limit; real workload is ≤ 0.5 A 3V3 typical (S3+H2+mux) = ~0.3 W dissipation, rise ≈ 24 °C. Well within SOA.

6. ESP32-S3 module — U5 ESP32-S3-WROOM-1-N16R8

Reference: Espressif ESP32-S3-WROOM-1/1U datasheet (N16R8 variant: 16 MB quad flash + 8 MB octal PSRAM).

Power

- 3V3 pins (module pin 2) → **3V3_RAIL**
- GND pins (module pins 1, 40, 41, EPAD) → **GND**
- EN (module pin 3) → **EN_S3_NODE** (10 kΩ **R_EN_S3** pullup to 3V3_RAIL + 1 µF **C_ESP_EN** to GND + SW1 RESET_S3 button to GND)

I/O used

GPIO	Module pin	Net	Destination
IO0	27	**BOOT_S3_BTN**	SW2 terminal 1 (module internal pullup)
IO4	4	**LED_WIFI_DRV**	R_LED_WIFI 1 kΩ → D3 anode
IO5	5	**LED_ACT_DRV**	R_LED_ACT 1 kΩ → D4 anode
IO6	6	**LED_ZB_DRV**	R_LED_ZB 1 kΩ → D5 anode
IO8	8	**I2C_SDA_3V3**	TCA9548A SDA
IO9	9	**I2C_SCL_3V3**	TCA9548A SCL
IO10	10	**MUX_RESET_N**	TCA9548A /RESET (+ 10 kΩ R_MUX_RST pullup to 3V3)
IO11	11	**S3_TO_H2_RST**	R_H2_RST_SER 1 kΩ → **H2_RESET_N** (H2 EN)
IO12	12	**H2_IRQ**	H2 IO10 (active low, H2 open-drain; 10 kΩ R_H2_IRQ pullup to 3V3)
IO13	13	**S3_TO_H2_BOOT**	R_H2_BOOT_SER 1 kΩ → **H2_BOOT_SEL** (H2 IO9 strap)
IO17	17	**H2_UART_TX**	H2 IO4 (H2's UART1 RX)
IO18	18	**H2_UART_RX**	H2 IO5 (H2's UART1 TX)
IO19	23	**USB_DN**	USBLC6 pins 1/6
IO20	24	**USB_DP**	USBLC6 pins 3/4
IO43	37	**U0TXD_S3**	JP_DEBUG_S3 pin 3, TP_U0TXD_S3
IO44	36	**U0RXD_S3**	JP_DEBUG_S3 pin 4, TP_U0RXD_S3

Module pin numbers per Espressif ESP32-S3-WROOM-1 datasheet rev 1.4+.

Reserved / no-copper-escape

- IO3, IO45, IO46: strap pins. No copper on any layer beyond module pad.
- IO26..IO32: internal SPI flash (N16 variant). Module-internal, no external connection.
- IO33..IO37: internal octal PSRAM (R8 variant). Module-internal, no external connection.
- IO48: NC (optional RGB LED on some module revs).

Decoupling (MANDATORY — place per Espressif reference design)

- **C_ESP_BULK** 22 µF 25 V X5R 0805 on 3V3_RAIL within 10 mm of module.

- At each module 3V3 entry (module has one 3V3 pin but internal distribution creates multiple VDD domains), three local decoupling pairs around module perimeter:
  - **C_ESP_DEC1a** 100 nF 0603 X7R + **C_ESP_DEC1b** 1 µF 0603 X5R
  - **C_ESP_DEC2a** 100 nF 0603 X7R + **C_ESP_DEC2b** 1 µF 0603 X5R
  - **C_ESP_DEC3a** 100 nF 0603 X7R + **C_ESP_DEC3b** 1 µF 0603 X5R
- **C_ESP_EN** 1 µF 25 V X5R 0603 from EN_S3_NODE to GND.

Rationale: WiFi burst transient current up to 500 mA requires paired bulk + HF local decoupling at each VDD entry.

7. ESP32-H2 module — U_H2 ESP32-H2-MINI-1-N4

Reference: Espressif ESP32-H2-MINI-1 datasheet (N4 variant: 4 MB SPI flash). RISC-V @ 96 MHz, IEEE 802.15.4 + Bluetooth LE 5.

Power

- 3V3 pins (module pin 2) → **3V3_RAIL**
- GND pins (module pins 1, 3, 25, 26, 27, 28/EPAD per datasheet) → **GND**
- EN (module pin 24) → **H2_RESET_N** (10 kΩ **R_EN_H2** pullup to 3V3_RAIL + 1 µF **C_H2_EN** to GND). Driven low by S3 IO11 (through 1 kΩ R_H2_RST_SER) during reset.

I/O used

H2 GPIO	Module pin	Direction	Net	Destination
IO2	20	Input, strap — tie HIGH	—	10 kΩ R_H2_STRAP2 → 3V3_RAIL
IO4	4	Input (H2 UART1 RX)	**H2_UART_TX**	S3 IO17
IO5	5	Output (H2 UART1 TX)	**H2_UART_RX**	S3 IO18
IO8	9	Input, strap — tie HIGH	—	10 kΩ R_H2_STRAP8 → 3V3_RAIL
IO9	10	Input, strap — **H2_BOOT_SEL**	**H2_BOOT_SEL**	10 kΩ R_H2_STRAP9 → 3V3_RAIL; R_H2_BOOT_SER 1 kΩ from S3 IO13; JP_DEBUG_H2 pin 6
IO10	11	Output, open-drain	**H2_IRQ**	S3 IO12 (10 kΩ R_H2_IRQ pullup to 3V3 on S3 side)
IO23	14	Output (H2 UART0 TX, debug)	**U0TXD_H2**	JP_DEBUG_H2 pin 3, TP_U0TXD_H2
IO24	15	Input (H2 UART0 RX, debug)	**U0RXD_H2**	JP_DEBUG_H2 pin 4, TP_U0RXD_H2
IO26	23	USB D− (H2 built-in USB	**H2_USB_DN**	TP_H2_USB_DN

		Serial/JTAG)		
IO27	22	USB D+ (H2 built-in USB Serial/JTAG)	**H2_USB_DP**	TP_H2_USB_DP

Module pin numbers per Espressif ESP32-H2-MINI-1 datasheet rev 1.1+.

Reserved

- IO0, IO1, IO3, IO11, IO12, IO13, IO14, IO22, IO25: unused — no copper escape beyond module pad.

Decoupling (MANDATORY)

- **C_H2_BULK** 10 µF 10 V X5R 0603 on 3V3_RAIL within 10 mm of module.
- Three local decoupling pairs adjacent to module 3V3 entry:
  - **C_H2_DEC1a** 100 nF 0603 X7R + **C_H2_DEC1b** 1 µF 0603 X5R
  - **C_H2_DEC2a** 100 nF 0603 X7R + **C_H2_DEC2b** 1 µF 0603 X5R
  - **C_H2_DEC3a** 100 nF 0603 X7R
- **C_H2_EN** 1 µF 25 V X5R 0603 from H2_RESET_N to GND.

H2 boot / programming strap

- Normal boot (SPI flash): H2 IO9 = HIGH. Achieved via R_H2_STRAP9 10 kΩ pullup to 3V3_RAIL. S3 IO13 held high-Z (input) during normal operation.
- UART download (for first flash or recovery): S3 IO13 driven LOW through R_H2_BOOT_SER 1 kΩ, overriding the 10 kΩ pullup at IO9 node. Then pulse H2_RESET_N low via S3 IO11 for ≥ 10 ms and release. H2 enters UART download mode. S3 drives esptool protocol on UART at 115200 bps (auto-baud).
- External factory flash alternative: JP_DEBUG_H2 pin 6 exposes H2_BOOT_SEL — external USB-UART dongle can assert low directly.

8. Reset and BOOT buttons

SW1 RESET_S3 — tactile SMD 6×6

- Terminal 1 → **EN_S3_NODE**
- Terminal 2 → **GND**

SW2 BOOT_S3 — tactile SMD 6×6

- Terminal 1 → **BOOT_S3_BTN**
- Terminal 2 → **GND**

No physical buttons for H2 — reset and boot controlled by S3 or JP_DEBUG_H2.

9. TCA9548A mux — U6 TCA9548APWR (TSSOP-24)

Power and address

Pin	Name	Net
24	VCC	**3V3_RAIL**
12	GND	**GND**
1	A0	**GND**

2	A1	**GND**
3	A2	**GND**

Resolved I2C address: **0x70**.

Master-side I2C

Pin	Name	Net	Pullup
22	SCL	**I2C_SCL_3V3**	4.7 kΩ **R_SCL_MCU** to 3V3_RAIL
23	SDA	**I2C_SDA_3V3**	4.7 kΩ **R_SDA_MCU** to 3V3_RAIL

Reset

- Pin 4 /RESET → **MUX_RESET_N** (S3 GPIO10). Pullup 10 kΩ **R_MUX_RST** to 3V3_RAIL.

Pin mapping (LOCKED — per TI SCPS195)

Pin	Name
1	A0
2	A1
3	A2
4	/RESET
5	SD0
6	SC0
7	SD1
8	SC1
9	SD2
10	SC2
11	SD3
12	GND
13	SC3
14	SD4
15	SC4
16	SD5
17	SC5
18	SD6
19	SC6
20	SD7
21	SC7
22	SCL
23	SDA
24	VCC

Downstream channel net assignments

Ch	SDA pin / net	SCL pin / net
0	pin 5 → **MUX_SDA0**	pin 6 → **MUX_SCL0**
1	pin 7 → **MUX_SDA1**	pin 8 → **MUX_SCL1**
2	pin 9 → **MUX_SDA2**	pin 10 → **MUX_SCL2**
3	pin 11 → **MUX_SDA3**	pin 13 → **MUX_SCL3**
4	pin 14 → **MUX_SDA4**	pin 15 → **MUX_SCL4**
5	pin 16 → **MUX_SDA5**	pin 17 → **MUX_SCL5**
6	pin 18 → **MUX_SDA6**	pin 19 → **MUX_SCL6**

7	pin 20 → **MUX_SDA7**	pin 21 → **MUX_SCL7**

Each MUX_SDA{n} / MUX_SCL{n}: 4.7 kΩ pullup to 3V3_RAIL (**R_SDA_MUX{n}** / **R_SCL_MUX{n}**).

Decoupling

- **C_MUX_DEC** 100 nF 0603 X7R between pin 24 and GND, adjacent to U6.

Debug header (pre-mux logic analyzer access) — JP_DEBUG_I2C

1×4 pin header 2.54 mm:

Pin	Net	Notes
1	**3V3_RAIL**	power
2	**GND**	
3	**I2C_SDA_3V3** (via 220 Ω **R_DBG_SDA**)	series resistor protects MCU from probe loading
4	**I2C_SCL_3V3** (via 220 Ω **R_DBG_SCL**)	

10. Per-row level shift + protection (identical block × 8, n = 1..8)

PCA9306DCTR — U{6+n} VSSOP-8

Pin	Name	Net
1	EN	**3V3_RAIL**
2	VREF1	**3V3_RAIL**
3	SDA1	**MUX_SDA{n-1}**
4	GND	**GND**
5	SCL1	**MUX_SCL{n-1}**
6	SCL2	**ROW{n}_SCL_5V**
7	VREF2	**5V_RAIL**
8	SDA2	**ROW{n}_SDA_5V**

Package selection (LOCKED)

- **VSSOP-8 (DCT suffix, PCA9306DCTR)** only. SOT-23-6 `PCA9306DP1R` variant MUST NOT substitute — different pinout, incompatible footprint.
- Pinout above matches TI SCPS169 VSSOP-8 datasheet exclusively.

5 V-side pullups

- **R_SDA_ROW{n}** 4.7 kΩ: **ROW{n}_SDA_5V** → **5V_RAIL**
- **R_SCL_ROW{n}** 4.7 kΩ: **ROW{n}_SCL_5V** → **5V_RAIL**

ESD array — U{14+n} PESD3V3L5UW (SOT-363)

Connected on the row-side of the level shifter, upstream of the connector: - Channel 1 (pins 1–6): **ROW{n}_SDA_5V** to **GND** - Channel 2 (pins 2–5): **ROW{n}_SCL_5V** to **GND** - Channel 3 (pins 3–4): tied to GND or left NC per Nexperia datasheet.

Power protection

- **F{n}** 2 A polyfuse 1812 (Littelfuse 1812L200/24THMR): between **5V_RAIL** and **ROW{n}_5V_OUT**.
- **D_ROW{n}** SMBJ5.0A unidirectional TVS (600 W, 5 V working, 9.2 V clamp): between **ROW{n}_5V_OUT** and **GND**.
- **C_ROW{n}** 47 µF 10 V X5R 1210: between **ROW{n}_5V_OUT** and **GND** at connector.
- **C_ROW{n}_HF** 100 nF 0603 X7R: between **ROW{n}_5V_OUT** and **GND** at connector.

Row connector — J{2+n} B4B-XH-A (4-pin THT, 2.5 mm pitch)

Pin	Signal
1	**ROW{n}_5V_OUT**
2	**GND**
3	**ROW{n}_SDA_5V**
4	**ROW{n}_SCL_5V**

11. Status LEDs

Ref	Color	Driver	Series R	From	To
D2	red	always-on	1 kΩ **R_LED_PWR**	3V3_RAIL	GND
D3	blue	S3 GPIO4	1 kΩ **R_LED_WIFI**	**LED_WIFI_DRV**	GND (cathode)
D4	green	S3 GPIO5	1 kΩ **R_LED_ACT**	**LED_ACT_DRV**	GND (cathode)
D5	yellow	S3 GPIO6	1 kΩ **R_LED_ZB**	**LED_ZB_DRV**	GND (cathode)

12. Debug headers and test points

JP_DEBUG_S3 — 1×4 header 2.54 mm (UART0 breakout for S3)

Pin	Net	Notes
1	**3V3_RAIL**	power, 0.5 A max
2	**GND**	
3	**U0TXD_S3** (S3 IO43)	S3 TX
4	**U0RXD_S3** (S3 IO44)	S3 RX

Primary S3 debug is the USB-C connector (native USB Serial/JTAG via S3 GPIO19/20). JP_DEBUG_S3 is secondary/bench access.

JP_DEBUG_H2 — 1×6 header 2.54 mm (UART0 + EN + BOOT for H2)

Pin	Net	Notes
1	**3V3_RAIL**	power, 0.3 A max
2	**GND**	
3	**U0TXD_H2** (H2 IO24)	H2 TX
4	**U0RXD_H2** (H2 IO23)	H2 RX
5	**H2_RESET_N** (H2	external dongle can

	EN)	assert low to reset
6	**H2_BOOT_SEL** (H2 IO9)	external dongle can assert low for UART download

Wiring compatible with Espressif ESP-Prog and FTDI-type USB-UART dongles in reset+boot auto-control mode.

H2 USB Serial/JTAG test pads

H2's built-in USB 2.0 Full-Speed peripheral on IO26/IO27 is not routed to a connector. Expose as 1.0 mm pads:

TP	Net
TP_H2_USB_DN	H2_USB_DN (H2 IO26)
TP_H2_USB_DP	H2_USB_DP (H2 IO27)

Access via wire-solder pigtail or pogo-pin jig for direct JTAG debug of H2 without going through S3.

Test points (all 1.0 mm exposed round pad, no drill, silkscreen label, L1 access)

Label	Net
TP_3V3	3V3_RAIL
TP_5V	5V_RAIL
TP_5V_LOGIC	5V_LOGIC
TP_GND	GND
TP_MCU_SDA	I2C_SDA_3V3
TP_MCU_SCL	I2C_SCL_3V3
TP_MUX_RST	MUX_RESET_N
TP_U0TXD_S3	U0TXD_S3
TP_U0RXD_S3	U0RXD_S3
TP_U0TXD_H2	U0TXD_H2
TP_U0RXD_H2	U0RXD_H2
TP_H2_UART_TX	H2_UART_TX
TP_H2_UART_RX	H2_UART_RX
TP_H2_RST	H2_RESET_N
TP_H2_IRQ	H2_IRQ
TP_H2_BOOT	H2_BOOT_SEL
TP_H2_USB_DP	H2_USB_DP
TP_H2_USB_DN	H2_USB_DN
TP_MUX_SDA0..7	MUX_SDA0..7
TP_MUX_SCL0..7	MUX_SCL0..7
TP_ROW1..8_5V	ROW1..8_5V_OUT
TP_ROW1..8_SDA	ROW1..8_SDA_5V
TP_ROW1..8_SCL	ROW1..8_SCL_5V

13. Mux bypass path

Zero-ohm solder bridges from the S3 I2C bus to TCA9548A channel 0 outputs:

- **JP_BYPASS_SDA** 0 Ω 0402, DNP by default — bridges **I2C_SDA_3V3** → **MUX_SDA0** when populated.
- **JP_BYPASS_SCL** 0 Ω 0402, DNP by default — bridges **I2C_SCL_3V3** → **MUX_SCL0** when populated.

Purpose: isolate TCA9548A failure from PCA9306 failure from cable failure during bring-up.

Operation: populate both jumpers AND hold MUX_RESET_N low (test clip) to force the TCA9548A high-Z. S3 then drives PCA9306 U7 directly → ROW1 XH-4.

Default state: **DNP**. Only populate for diagnostic runs. Remove before deployment.

14. Inter-MCU UART link — specification

Parameter	Value
Physical layer	CMOS 3.3 V logic, push-pull, single-ended
Nets	H2_UART_TX (S3→H2), H2_UART_RX (H2→S3), H2_RESET_N (S3→H2), H2_IRQ (H2→S3, open-drain), H2_BOOT_SEL (S3→H2, strap override)
Baud rate	**921 600 bps**
Data format	8 data bits, no parity, 1 stop bit (8N1)
Flow control	**None** (hardware RTS/CTS not used; protocol-level flow control via ACK/NACK and bounded-buffer credits)
Framing	**COBS** (Consistent Overhead Byte Stuffing) with 0x00 delimiter byte
Integrity	**CRC-16/CCITT-FALSE** (poly 0x1021, init 0xFFFF, no reflect, xorout 0x0000) on frame payload
Roles	S3 is MASTER. H2 is RESPONDER.
Transaction model	Command/response. Every S3 command expects an H2 reply (ACK, NACK, or data).
H2_IRQ semantics	H2 asserts low when it has asynchronous data (e.g. inbound Zigbee message). S3 responds by issuing a POLL command.
Max frame length	256 bytes payload (before COBS + CRC).
S3 UART peripheral	UART1
H2 UART peripheral	UART1 (IO4=RX, IO5=TX — GPIO matrix routed)
H2 UART0 (IO23/24)	Reserved for H2 ROM bootloader + debug console at 115200 8N1, via JP_DEBUG_H2 only

Hard rule: no SPI between S3 and H2. Capture ERC must enforce that no net named `*_SPI_*`, `*_MISO*`, `*_MOSI*`, `*_SCK*`, `*_CS_H2*` exists.

15. Power summary

Net	Source	Load
USB_5V_IN	USB-C VBUS via FB1	U1 pin 5, U2 pin 1, C_USB_IN
		5V_RAIL (direct), D_BACKFEED anode,

JACK_5V	barrel via Q1	C_BULK_INPUT, C_JACK_IN
5V_LOGIC	U2 LM66100 OR D_BACKFEED	U4 pin 2 (buck in), C_LOGIC_IN
5V_RAIL	JACK_5V (direct)	F1..F8, PCA9306 VREF2 (×8), C_BULK_5V_RAIL, C_BULK_5V_CER, C_BULK_5V_HF
3V3_RAIL	U4 AP63300 + L_BUCK	S3 VDD, H2 VDD, TCA9548A VCC, PCA9306 VREF1 + EN (×8), all 3V3-side pullups, power LED, S3/H2 EN pullups
GND	common return	all

Maximum system input current (spec): **8 A** total at 5V_RAIL via barrel jack. Per-row polyfuse: 2 A hold. 3V3 current budget: ~800 mA peak (S3 500 mA WiFi burst + H2 150 mA BLE/Zigbee TX + 150 mA headroom). Buck rated 3 A — 3.7× margin.

16. I2C pullup summary

Net	Master-side R	Row-side R	Endpoint A	Endpoint B
I2C_SDA_3V3	4.7 kΩ to 3V3	—	S3 GPIO8	TCA9548A SDA
I2C_SCL_3V3	4.7 kΩ to 3V3	—	S3 GPIO9	TCA9548A SCL
MUX_SDA0..7	4.7 kΩ to 3V3	—	TCA9548A ch n SDA	PCA9306 SDA1
MUX_SCL0..7	4.7 kΩ to 3V3	—	TCA9548A ch n SCL	PCA9306 SCL1
ROW{n}_SDA_5V	—	4.7 kΩ to 5V	PCA9306 SDA2	J{2+n} pin 3 via PESD
ROW{n}_SCL_5V	—	4.7 kΩ to 5V	PCA9306 SCL2	J{2+n} pin 4 via PESD

Firmware rule: mux channel-select register accepts only single-bit values (0x01, 0x02, 0x04, 0x08, 0x10, 0x20, 0x40, 0x80). Multi-bit masks are rejected — parallel row pullups would exceed ATmega328P I_OL.

17. Package and pin-1 reference (BOM validation)

Ref	Part	Package	Pin-1 marker	Authoritat datashe
U1	USBLC6-2SC6	SOT-23-6 / SC-74-6	Dot, top-left with part number readable L-to-R. Pin 1 = I/O 1 (D−). Pin 4 tied to pin 3 (I/O 2), **NOT ground**. Pin 1 = IN, pin 5 = OUT.	ST DS3019

U2	LM66100DCKR	SOT-23-5	Single instance; USB path only.	TI SNVSBL1
U4	AP63300WU-7	TSOT-26 (W suffix)	Dot on pin 1 corner. Pin 1 = SW, pin 6 = BST.	Diodes AP63300 datasheet r 2+
U5	ESP32-S3-WROOM-1-N16R8	65-pin SMD module (18 × 25.5 mm)	Marking on module silk; pin 1 at antenna-opposite corner.	Espressif ESP32-S3-WROOM-1/ datasheet r 1.4+
U_H2	ESP32-H2-MINI-1-N4	28-pad SMD module (13.2 × 16.6 mm)	Marking on module silk.	Espressif ESP32-H2-MINI-1 datasheet r 1.1+
U6	TCA9548APWR	TSSOP-24 PW	Dot/dimple on one narrow end; pin 1 = A0. **Trap**: RGE (QFN) variant has different pad pattern. Order **only PWR suffix**.	TI SCPS195
U7..U14	PCA9306DCTR	VSSOP-8 (DCT suffix)	Dot on pin 1 corner; pin 1 = EN. **Trap**: `PCA9306DP1R` is SOT-23-6 with different pinout — do not substitute.	TI SCPS169
U15..U22	PESD3V3L5UW	SOT-363 (SC-88)	Pin 1 marker per Nexperia datasheet.	Nexperia PESD3V3L5 datasheet
Q1	AO3401A	SOT-23-3	Gate = pin 1, source = pin 3, drain = pin 2.	AOS AO340 datasheet
D1	SMAJ5.0CA	DO-214AC (SMA)	CA suffix = bidirectional, no polarity.	Littelfuse S series datasheet
D_ROW1..8	SMBJ5.0A	DO-214AA (SMB)	A suffix = unidirectional; cathode band to +5 V.	Littelfuse S series datasheet
D_BACKFEED	SS14	DO-214AC (SMA)	Cathode band on V_OUT side; anode on JACK_5V, cathode on 5V_LOGIC.	ON Semi / Vishay SS1 datasheet

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CRITICAL DESIGN RULES (MUST FOLLOW)

1. **High-current row power MUST NOT pass through LM66100.** JACK_5V → 5V_RAIL direct; LM66100 (U2) is on USB path only.
2. **Inter-MCU communication MUST be UART only.** SPI between S3 and H2 is forbidden. ERC enforces this.
3. **Buck regulator layout (§5) MUST follow AP63300 datasheet.** SW node ≤ 3 mm², L_BUCK ≤ 4 mm from U4, C_BOOT ≤ 3 mm from U4 between pins 1 (SW) and 6 (BST).
4. **All IC pinouts MUST be verified against vendor datasheets at capture time.** No approximate pin mappings — §6, §7, §9, §10 tables are authoritative.
5. **5V_RAIL (row power) and 5V_LOGIC (regulator input) are distinct nets.** Do not short them on PCB.
6. **D_BACKFEED Schottky is the only connection between JACK_5V and 5V_LOGIC.** Polarity: anode JACK_5V, cathode 5V_LOGIC. Reversal back-drives the buck from logic into row power.
7. **Antenna keep-outs are hard layout rules.** 15×7 mm under S3 antenna, 11×6 mm under H2 antenna, cleared on all 4 layers. Modules on opposite board edges. ≥ 30 mm antenna edge-to-edge separation.
8. **H2 is a slave.** S3 is the sole system controller. H2 cannot drive MUX, row I2C, or power state.

Part 3 — PINOUT

Master v2 — GPIO pinouts

Rev B. Full GPIO allocation for both MCUs.

ESP32-S3-WROOM-1-N16R8 (primary)

GPIO	Module pin	Function	Net	Direction	
IO0	27	Strap + BOOT button	**BOOT_S3_BTN**	Input (module pullup)	Active during → ROM Trace : GND-fl
IO1	13	reserved	—	—	Free fo
IO2	14	reserved	—	—	Free fo
IO3	15	Strap pin	—	—	**No co escap**
IO4	4	LED	**LED_WIFI_DRV**	Output	1 kΩ – GND. I WiFi st
IO5	5	LED	**LED_ACT_DRV**	Output	1 kΩ – → GND I2C/we pulse
IO6	6	LED	**LED_ZB_DRV**	Output	1 kΩ – → GND Zigbee
IO7	7	reserved	—	—	Free fo
IO8	8	I2C SDA	**I2C_SDA_3V3**	Open-drain bidir	4.7 kΩ 3V3. T TCA95
IO9	9	I2C SCL	**I2C_SCL_3V3**	Open-drain bidir	4.7 kΩ 3V3. T TCA95
IO10	10	MUX reset	**MUX_RESET_N**	Output, active low	10 kΩ 3V3. D

		reset		active low	TCA95
IO11	11	H2 reset	**S3_TO_H2_RST → H2_RESET_N**	Output, active low	1 kΩ s (R_H2_ Extern pullup Defaul during pulse H2.
IO12	12	H2 IRQ	**H2_IRQ**	Input, active low	H2 op output pullup assert it has
IO13	13	H2 boot select	**S3_TO_H2_BOOT → H2_BOOT_SEL**	Output	1 kΩ s (R_H2_ Extern pullup Defaul drive H2 in downl for OT
IO14	14	reserved	—	—	Free fo
IO15	15	reserved	—	—	Free fo
IO16	16	reserved	—	—	Free fo
IO17	17	H2 UART1 TX	**H2_UART_TX**	Output	92160 To H2
IO18	18	H2 UART1 RX	**H2_UART_RX**	Input	From H
IO19	23	USB D−	**USB_DN**	Bidir	To USB 1/6. 90 with IO
IO20	24	USB D+	**USB_DP**	Bidir	To USB 3/4. 90 with IO
IO21	25	reserved	—	—	Free fo
IO26–IO32	—	internal	—	—	SPI fla Modul
IO33–IO37	—	internal	—	—	Octal Modul
IO38	33	reserved	—	—	Free fo
IO39	34	reserved	—	—	Free fo
IO40	35	reserved	—	—	Free fo
IO41	26	reserved	—	—	Free fo
IO42	28	reserved	—	—	Free fo
IO43	37	UART0 TX	**U0TXD_S3**	Output	To JP_D pin 3 - TP_U0 ROM b
IO44	36	UART0 RX	**U0RXD_S3**	Input	To JP_D pin 4 - TP_U0
IO45	29	Strap pin	—	—	**No co escap**
IO46	30	Strap pin	—	—	**No co escap**
IO47	31	reserved	—	—	Free fo
					NC (sc

IO48	32	reserved	—	—	NC (so have c LED)
EN	3	Chip enable + reset	**EN_S3_NODE**	Input	10 kΩ 3V3, 1 SW1 R button
VDD3V3	2	3.3 V supply	**3V3_RAIL**	Power	Decou of SCH
GND	1, 40, 41, EPAD	Ground	**GND**	Return	Stitch 9 ther L2

Module pin numbers per Espressif ESP32-S3-WROOM-1 datasheet.

S3 strap-pin rules

- **IO0**: boot mode. Module integrates pullup. BOOT_S3 button pulls to GND. Strap: high = normal SPI flash boot, low at reset = download boot.
- **IO3, IO45, IO46**: no copper connection on any layer beyond the module pad. Retains module-default behaviour.

S3 firmware-visible constraints

- I2C on GPIO8/9: routed through GPIO matrix. Arduino: `Wire.begin(8, 9);`.
- MUX_RESET_N (GPIO10): active low, ≥ 10 µs pulse during bus-stuck recovery. Keep high in normal operation.
- H2_RESET_N / H2_BOOT_SEL: configured as inputs (high-Z) during normal operation; switched to outputs only for H2 reset or OTA entry. 1 kΩ series resistors protect against contention with JP_DEBUG_H2 external dongle.
- H2_IRQ: interrupt-capable input, falling-edge trigger, internal weak pull-up disabled (external 10 kΩ handles it).
- H2_UART_TX/RX on IO17/IO18: mapped to UART1 via GPIO matrix. 921600 bps 8N1.
- LEDs on IO4/5/6: active high, direct drive through 1 kΩ.
- UART0 (IO43/44): boot-log + debug console. JP_DEBUG_S3 + test pads.

ESP32-H2-MINI-1-N4 (radio coprocessor)

H2 GPIO	Module pin	Function	Net	Directio
EN	24	Chip enable	**H2_RESET_N**	Input
IO0	7	Strap (reserved)	—	—
IO1	8	reserved	—	—

IO2	20	**Strap (ROM_MSG)**	**H2_STRAP_IO2**	Input (strap)
IO3	21	reserved	—	—
IO4	4	UART1 RX (from S3)	**H2_UART_TX**	Input
IO5	5	UART1 TX (to S3)	**H2_UART_RX**	Output
IO8	9	**Strap (ROM_MSG_PRINT)**	**H2_STRAP_IO8**	Input (strap)
IO9	10	**Strap (BOOT_MODE) + S3 override**	**H2_BOOT_SEL**	Input (strap)
IO10	11	IRQ to S3	**H2_IRQ**	Output, open-drain
IO11	12	reserved	—	—
IO12	16	reserved	—	—
IO13	17	reserved	—	—
IO14	18	reserved	—	—
IO22	13	reserved	—	—
IO23	14	UART0 RX (debug)	**U0RXD_H2**	Input
IO24	15	UART0 TX (debug)	**U0TXD_H2**	Output
IO25	19	reserved	—	—

IO26	23	USB D− (USB Serial/JTAG)	**H2_USB_DN**	Bidir
IO27	22	USB D+ (USB Serial/JTAG)	**H2_USB_DP**	Bidir
VDD3V3	2	3.3 V supply	**3V3_RAIL**	Power
GND	1, 3, 25, 26, 27, EPAD	Ground	**GND**	Return

Module pin numbers per Espressif ESP32-H2-MINI-1 datasheet.

H2 strap-pin rules

- **IO2**: tied high via R_H2_STRAP2 (10 kΩ to 3V3). Strap sampled at EN release.
- **IO8**: tied high via R_H2_STRAP8 (10 kΩ to 3V3). High = ROM messages enabled on U0TXD_H2 at boot.
- **IO9**: pulled high via R_H2_STRAP9 (10 kΩ to 3V3). Default boot = SPI flash. S3 IO13 can override to low through 1 kΩ series for UART download boot (OTA update of H2 firmware from S3). JP_DEBUG_H2 pin 6 exposes the same net for external dongle access.

H2 firmware-visible constraints

- UART1 (IO4/IO5) carries the S3 command/response protocol. 921600 bps 8N1, COBS + CRC-16/CCITT-FALSE framing. H2 is responder only.
- UART0 (IO23/IO24): ROM bootloader + debug log. Accessible via JP_DEBUG_H2 only.
- H2_IRQ (IO10): driven as open-drain output. Default high-Z (high via S3 pullup). Firmware pulls low to signal pending async data.
- 802.15.4 channels 11–26: recommended pin to channel 25 or 26 to minimize WiFi co-channel interference with S3. Application-layer constraint — see coexistence note in README.md.
- USB Serial/JTAG on IO26/27: enabled by default in ROM. Allows direct programming of H2 via TP_H2_USB_DP/DN without going through S3. Independent of JP_DEBUG_H2 UART path.

Inter-MCU signal summary

Signal	Driver	Receiver	Default	Purpose
H2_UART_TX	S3 IO17	H2 IO4	—	Command frames from S3 to H2
H2_UART_RX	H2 IO5	S3 IO18	—	Response frames from H2 to S3
H2_RESET_N	S3 IO11 (1 kΩ series)	H2 EN	High (10 kΩ pullup)	S3 resets H2
	H2 IO10		High (10	H2 signals

H2_IRQ	(open-drain)	S3 IO12	kΩ pullup)	async event to S3
H2_BOOT_SEL	S3 IO13 (1 kΩ series)	H2 IO9	High (10 kΩ pullup)	S3 puts H2 in download mode for OTA

5 signals total between the two MCUs. No more, no less. No SPI.

Reserved-for-expansion GPIO list (S3)

S3 free GPIO for future expansion headers or features:

- IO1, IO2 (general use)
- IO7 (general use)
- IO14, IO15, IO16 (general use)
- IO21 (general use)
- IO38, IO39, IO40, IO41, IO42 (general use; no strap, no reserved role)
- IO47, IO48 (general use; IO48 may be RGB on some module revs — check before use)

Expansion header not fitted in Rev B. If added in future, prefer IO38–42.

Part 4 — BOM

Designator	Value	Package	Quantity	Mar
U1	USBLC6-2SC6	SOT-23-6	1	USBLC
U2	LM66100DCKR	SOT-23-5	1	LM661
U4	AP63300WU-7	TSOT-26	1	AP633
U5	ESP32-S3-WROOM-1-N16R8	ESP32-S3-WROOM-1	1	ESP32 N16R8
U_H2	ESP32-H2-MINI-1-N4	ESP32-H2-MINI-1	1	ESP32
U6	TCA9548APWR	TSSOP-24	1	TCA95
U7	PCA9306DCTR	VSSOP-8	1	PCA93
U8	PCA9306DCTR	VSSOP-8	1	PCA93
U9	PCA9306DCTR	VSSOP-8	1	PCA93
U10	PCA9306DCTR	VSSOP-8	1	PCA93
U11	PCA9306DCTR	VSSOP-8	1	PCA93
U12	PCA9306DCTR	VSSOP-8	1	PCA93
U13	PCA9306DCTR	VSSOP-8	1	PCA93
U14	PCA9306DCTR	VSSOP-8	1	PCA93

U14	PCA9306DCUR	VSSOP-8	1	PCA9
U15	PESD3V3L5UW	SOT-363	1	PESD3
U16	PESD3V3L5UW	SOT-363	1	PESD3
U17	PESD3V3L5UW	SOT-363	1	PESD3
U18	PESD3V3L5UW	SOT-363	1	PESD3
U19	PESD3V3L5UW	SOT-363	1	PESD3
U20	PESD3V3L5UW	SOT-363	1	PESD3
U21	PESD3V3L5UW	SOT-363	1	PESD3
U22	PESD3V3L5UW	SOT-363	1	PESD3
Q1	AO3401A	SOT-23-3	1	AO340
L_BUCK	4.7uH 3A shielded	SMD 6x6	1	MWSA
D1	SMAJ5.0CA	DO-214AC (SMA)	1	SMAJ5
D_BACKFEED	SS14 Schottky 40V 1A	DO-214AC (SMA)	1	SS14
D2	LED Red 0603	LED_0603	1	KT-060
D3	LED Blue 0603	LED_0603	1	KT-060
D4	LED Green 0603	LED_0603	1	KT-060
D5	LED Yellow 0603	LED_0603	1	KT-060
D_ROW1	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW2	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW3	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW4	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW5	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW6	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW7	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
D_ROW8	SMBJ5.0A	DO-214AA (SMB)	1	SMBJ5
FB1	Ferrite 470 ohm @100MHz	0603	1	BLM18
F1	Polyfuse 2A hold	1812	1	1812L
F2	Polyfuse 2A hold	1812	1	1812L
F3	Polyfuse 2A hold	1812	1	1812L
F4	Polyfuse 2A hold	1812	1	1812L
F5	Polyfuse 2A hold	1812	1	1812L
F6	Polyfuse 2A hold	1812	1	1812L
F7	Polyfuse 2A hold	1812	1	1812L

F8	Polyfuse 2A hold	1812	1	1812L
J1	USB-C receptacle 16-pin	USB-C_16P	1	USB41
J2	DC jack 5.5x2.1mm	DC-005	1	DC-00
J3	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J4	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J5	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J6	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J7	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J8	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J9	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
J10	B4B-XH-A(LF)(SN)	XH-4 THT	1	B4B-XI
JP_DEBUG_I2C	1x4 header 2.54mm	HDR-1X4	1	2211S
JP_DEBUG_S3	1x4 header 2.54mm	HDR-1X4	1	2211S
JP_DEBUG_H2	1x6 header 2.54mm	HDR-1X6	1	2211S
JP_BYPASS_SDA	0R 0402 DNP	0402	1	0402W
JP_BYPASS_SCL	0R 0402 DNP	0402	1	0402W
SW1	Tactile SMD 6x6	TACT_SMD_6x6	1	TS-118
SW2	Tactile SMD 6x6	TACT_SMD_6x6	1	TS-118
R_CC1	5.1k 1%	0603	1	0603W
R_CC2	5.1k 1%	0603	1	0603W
R_QGATE	10k 1%	0603	1	0603W
R_CE_USB	100k 1%	0603	1	0603W
R_EN_BUCK	100k 1%	0603	1	0603W
R_FB_TOP	100k 1%	0603	1	0603W
R_FB_BOT	31.6k 1%	0603	1	0603W
R_EN_S3	10k 1%	0603	1	0603W
R_EN_H2	10k 1%	0603	1	0603W
R_MUX_RST	10k 1%	0603	1	0603W
R_H2_RST_SER	1k 1%	0603	1	0603W
R_H2_BOOT_SER	1k 1%	0603	1	0603W

R_H2_IRQ	10k 1%	0603	1	0603W
R_H2_STRAP2	10k 1%	0603	1	0603W
R_H2_STRAP8	10k 1%	0603	1	0603W
R_H2_STRAP9	10k 1%	0603	1	0603W
R_SDA_MCU	4.7k 1%	0603	1	0603W
R_SCL_MCU	4.7k 1%	0603	1	0603W
R_SDA_MUX0	4.7k 1%	0603	1	0603W
R_SCL_MUX0	4.7k 1%	0603	1	0603W
R_SDA_MUX1	4.7k 1%	0603	1	0603W
R_SCL_MUX1	4.7k 1%	0603	1	0603W
R_SDA_MUX2	4.7k 1%	0603	1	0603W
R_SCL_MUX2	4.7k 1%	0603	1	0603W
R_SDA_MUX3	4.7k 1%	0603	1	0603W
R_SCL_MUX3	4.7k 1%	0603	1	0603W
R_SDA_MUX4	4.7k 1%	0603	1	0603W
R_SCL_MUX4	4.7k 1%	0603	1	0603W
R_SDA_MUX5	4.7k 1%	0603	1	0603W
R_SCL_MUX5	4.7k 1%	0603	1	0603W
R_SDA_MUX6	4.7k 1%	0603	1	0603W
R_SCL_MUX6	4.7k 1%	0603	1	0603W
R_SDA_MUX7	4.7k 1%	0603	1	0603W
R_SCL_MUX7	4.7k 1%	0603	1	0603W
R_SDA_ROW1	4.7k 1%	0603	1	0603W
R_SCL_ROW1	4.7k 1%	0603	1	0603W
R_SDA_ROW2	4.7k 1%	0603	1	0603W
R_SCL_ROW2	4.7k 1%	0603	1	0603W
R_SDA_ROW3	4.7k 1%	0603	1	0603W
R_SCL_ROW3	4.7k 1%	0603	1	0603W
R_SDA_ROW4	4.7k 1%	0603	1	0603W

R_SCL_ROW4	4.7k 1%	0603	1	0603W
R_SDA_ROW5	4.7k 1%	0603	1	0603W
R_SCL_ROW5	4.7k 1%	0603	1	0603W
R_SDA_ROW6	4.7k 1%	0603	1	0603W
R_SCL_ROW6	4.7k 1%	0603	1	0603W
R_SDA_ROW7	4.7k 1%	0603	1	0603W
R_SCL_ROW7	4.7k 1%	0603	1	0603W
R_SDA_ROW8	4.7k 1%	0603	1	0603W
R_SCL_ROW8	4.7k 1%	0603	1	0603W
R_DBG_SDA	220R 1%	0603	1	0603W
R_DBG_SCL	220R 1%	0603	1	0603W
R_SH1	1M 1%	0603	1	0603W
R_LED_PWR	1k 1%	0603	1	0603W
R_LED_WIFI	1k 1%	0603	1	0603W
R_LED_ACT	1k 1%	0603	1	0603W
R_LED_ZB	1k 1%	0603	1	0603W
C_FB	22pF C0G	0603	1	CL10C
C_BOOT	100nF X7R 25V	0603	1	CL10B
C_BUCK_IN	22uF 25V X5R	1210	1	CS322
C_BUCK_OUT_1	22uF 25V X5R	0805	1	CL21A
C_BUCK_OUT_2	22uF 25V X5R	0805	1	CL21A
C_BULK_INPUT	470uF 10V Al polymer	SMD 8x10.2	1	EEH-Z
C_BULK_5V_RAIL	220uF 10V Al polymer	SMD 8x7	1	EEH-Z
C_BULK_5V_CER	47uF 10V X5R	1210	1	CL32A
C_BULK_5V_HF	100nF X7R 25V	0603	1	CL10B
C_USB_IN	10uF 25V X7R	0805	1	CL21B
C_JACK_IN	10uF 25V X7R	0805	1	CL21B
C_LOGIC_IN	10uF 25V X7R	0805	1	CL21B
C_SH1	10nF X7R 25V	0603	1	CL10B
C_ESP_BULK	22uF 25V X5R	0805	1	CL21A
C_ESP_EN	1uF 25V X5R	0603	1	CL10A
C_ESP_DEC1a	100nF X7R 25V	0603	1	CL10B
C_ESP_DEC1b	1uF 25V X5R	0603	1	CL10A

C_ESP_DEC2a	100nF X7R 25V	0603	1	CL10B
C_ESP_DEC2b	1uF 25V X5R	0603	1	CL10A
C_ESP_DEC3a	100nF X7R 25V	0603	1	CL10B
C_ESP_DEC3b	1uF 25V X5R	0603	1	CL10A
C_H2_BULK	10uF 10V X7R	0603	1	CL10B
C_H2_EN	1uF 25V X5R	0603	1	CL10A
C_H2_DEC1a	100nF X7R 25V	0603	1	CL10B
C_H2_DEC1b	1uF 25V X5R	0603	1	CL10A
C_H2_DEC2a	100nF X7R 25V	0603	1	CL10B
C_H2_DEC2b	1uF 25V X5R	0603	1	CL10A
C_H2_DEC3a	100nF X7R 25V	0603	1	CL10B
C_MUX_DEC	100nF X7R 25V	0603	1	CL10B
C_ROW1	47uF 10V X5R	1210	1	CL32A
C_ROW2	47uF 10V X5R	1210	1	CL32A
C_ROW3	47uF 10V X5R	1210	1	CL32A
C_ROW4	47uF 10V X5R	1210	1	CL32A
C_ROW5	47uF 10V X5R	1210	1	CL32A
C_ROW6	47uF 10V X5R	1210	1	CL32A
C_ROW7	47uF 10V X5R	1210	1	CL32A
C_ROW8	47uF 10V X5R	1210	1	CL32A
C_ROW1_HF	100nF X7R 25V	0603	1	CL10B
C_ROW2_HF	100nF X7R 25V	0603	1	CL10B
C_ROW3_HF	100nF X7R 25V	0603	1	CL10B
C_ROW4_HF	100nF X7R 25V	0603	1	CL10B
C_ROW5_HF	100nF X7R 25V	0603	1	CL10B
C_ROW6_HF	100nF X7R 25V	0603	1	CL10B
C_ROW7_HF	100nF X7R 25V	0603	1	CL10B
C_ROW8_HF	100nF X7R 25V	0603	1	CL10B